

How Antivirus Works (Step-by-Step):

◆ **1. Scanning Files**

- It checks files on your computer using two methods:
 - **Real-time Scan:** Automatically scans when you open, download, or run a file.
 - **Full/Manual Scan:** Scans all files when you request it.

◆ **2. Virus Definitions (Signature-Based Detection)**

- Antivirus has a **database of known viruses**.
- It matches file code with virus **signatures** to detect threats.
 - 📌 Example: Think of this like matching a criminal's fingerprint in police records.

◆ **3. Heuristic Analysis**

- Detects **unknown viruses** by looking for suspicious behavior.
 - Example: A file trying to modify system files or send data secretly.

◆ **4. Behavior Monitoring**

- Watches how a program behaves while running.
- If a program acts like malware (e.g., encrypting files quickly), it's flagged or blocked.

◆ **5. Sandboxing (Advanced)**

- Runs suspicious files in a **safe virtual environment** to check behavior without risking your system.

◆ **6. Quarantine**

- If a threat is found, it's **isolated** so it can't harm the system.
- You can later choose to delete or restore the file.

◆ **7. Updates**

- Antivirus software **updates daily** to stay ahead of new threats.

What Antivirus Protects You From:

- Email scams (phishing)
- Infected USB drives
- Fake websites
- Malicious downloads
- Keyloggers (steal passwords)

Example:

1. You download a file from the internet.
2. Antivirus scans it immediately.
3. If it's safe → allowed to run.
 - If it's suspicious → blocked or quarantined.